

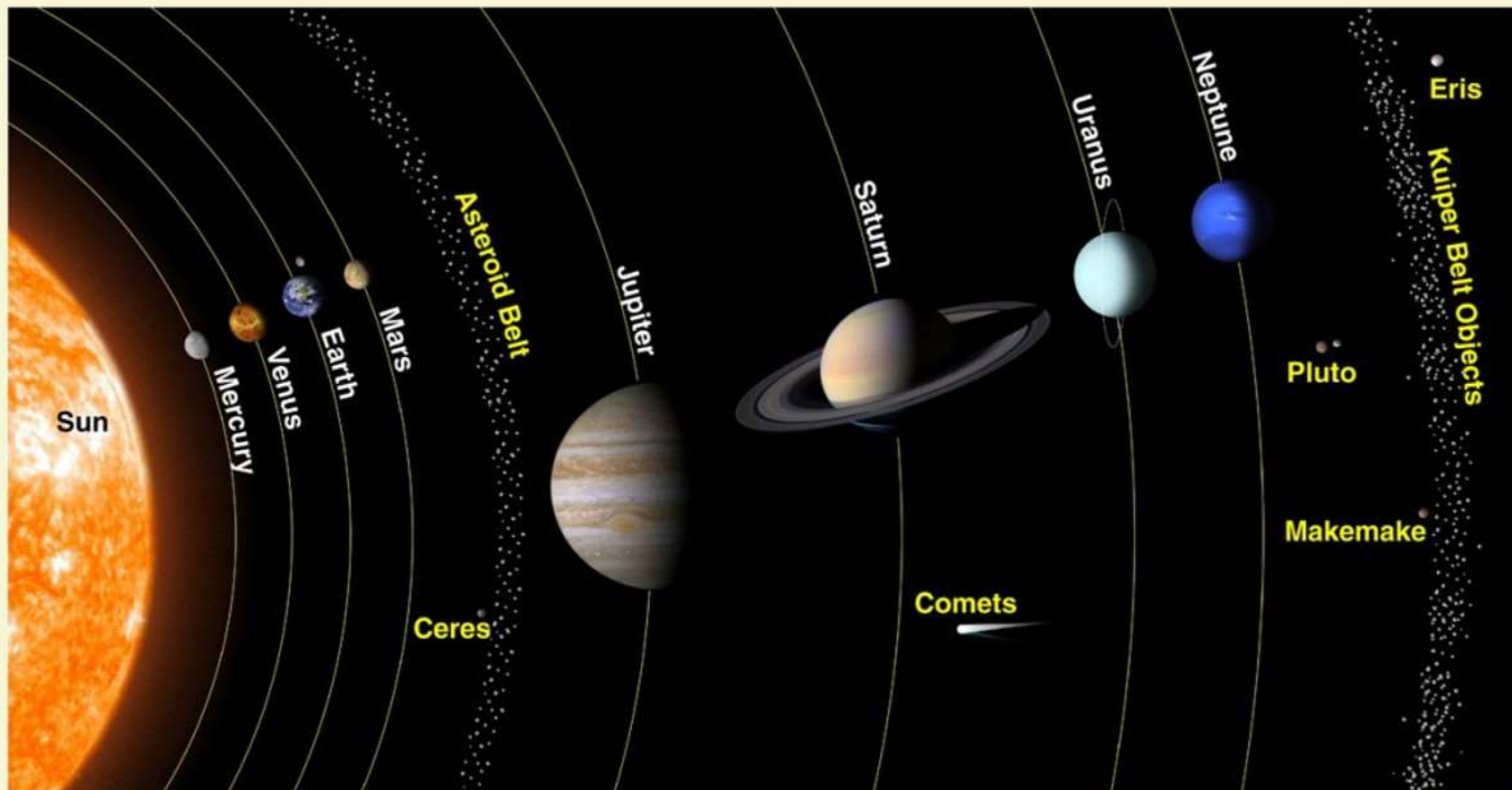
GEOGRAPHY

PHYSICAL GEOGRAPHY



MODULE -4

- WHY STARS TWINKLE?
- WHY VENUS IS HOTTER THAN MERCURY?
- WHY MARS IS CALLED RED PLANET?
- WHY OCEANS LOOK BLUE?



Mercury 8 th	Venus 6 th	Earth 5 th	Mars 7 th	Jupiter 1 st	Saturn 2 nd	Uranus 3 rd	Neptune 4 th
INNER PLANETS				OUTER PLANETS / JOVIAN PLANETS (99% mass)			
Closely spaced Rocky and Terrestrial Silicates and Metals				Gas Giants Hydrogen & Helium		Ice Giants Water, Ammonia & Methane	
No Moon	No Moon	1 Moon	2 Moons	69 Moons	61 Moons	27 Moons	14 Moons
Smallest	Hottest @462°C			Largest & Heaviest			
	Morning Star/ Evening Star	Supports Life	Red Planet	Has Rings	Has Rings	Has Rings	Has Rings
	Earth's Twin	Blue Planet	Highest Mountain (Olympus Mons)	Lord of the Heavens			
	Brightest Planet						
88 Days	225 Days	365 Days	688 Days	11.9 Years	29.5 Years	84 Years	164.8 Years

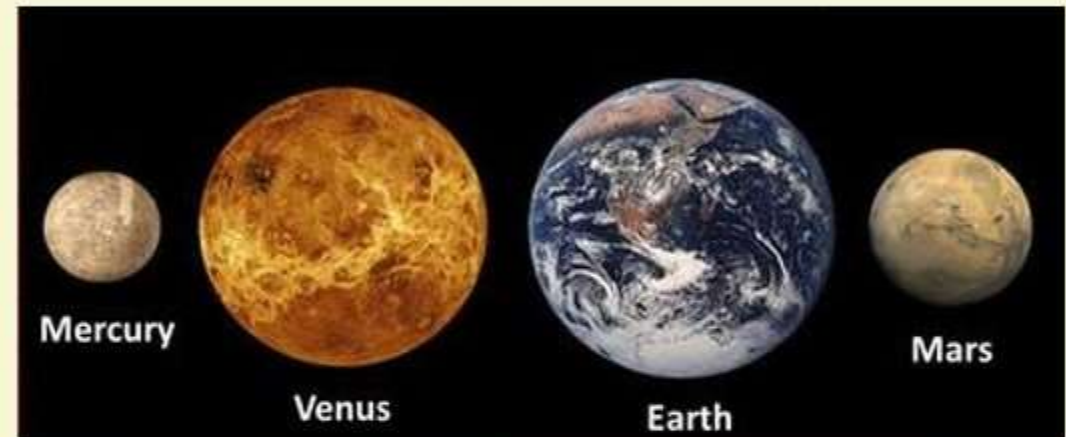
Inner and outer planets are separated by Asteroid belt





Why the planets closest to the Sun are rocky?

- The Sun was radiant enough to drive away gas from the inner solar system.
- Subsequently, it left behind the rocky cores which are now terrestrial planets.
- Close to the Sun, the temperature was too high for volatiles to condense. So, only the materials with higher melting point were able to form in this region.
- As the gas giants are away from the Sun, the temperature was cool enough for these volatile gases to condense and form these huge, less dense planets.





Why Mercury and Venus do not have moons?

- Most likely, because they are too close to the Sun.
- Any moon with too great a distance from these planets would be in an unstable orbit and be captured by the Sun.
- Sun is much larger than Mercury.
- It would most likely get sucked in the Sun's huge gravity.
- Mercury will never have the opportunity to capture smaller and less massive celestial bodies.
- Several planets obtained satellites by capturing asteroids and smaller celestial bodies.
- One example is Mars, Mars obtained two moons, Phobos and Deimos in this manner.



MOONS OF MARS

Phobos



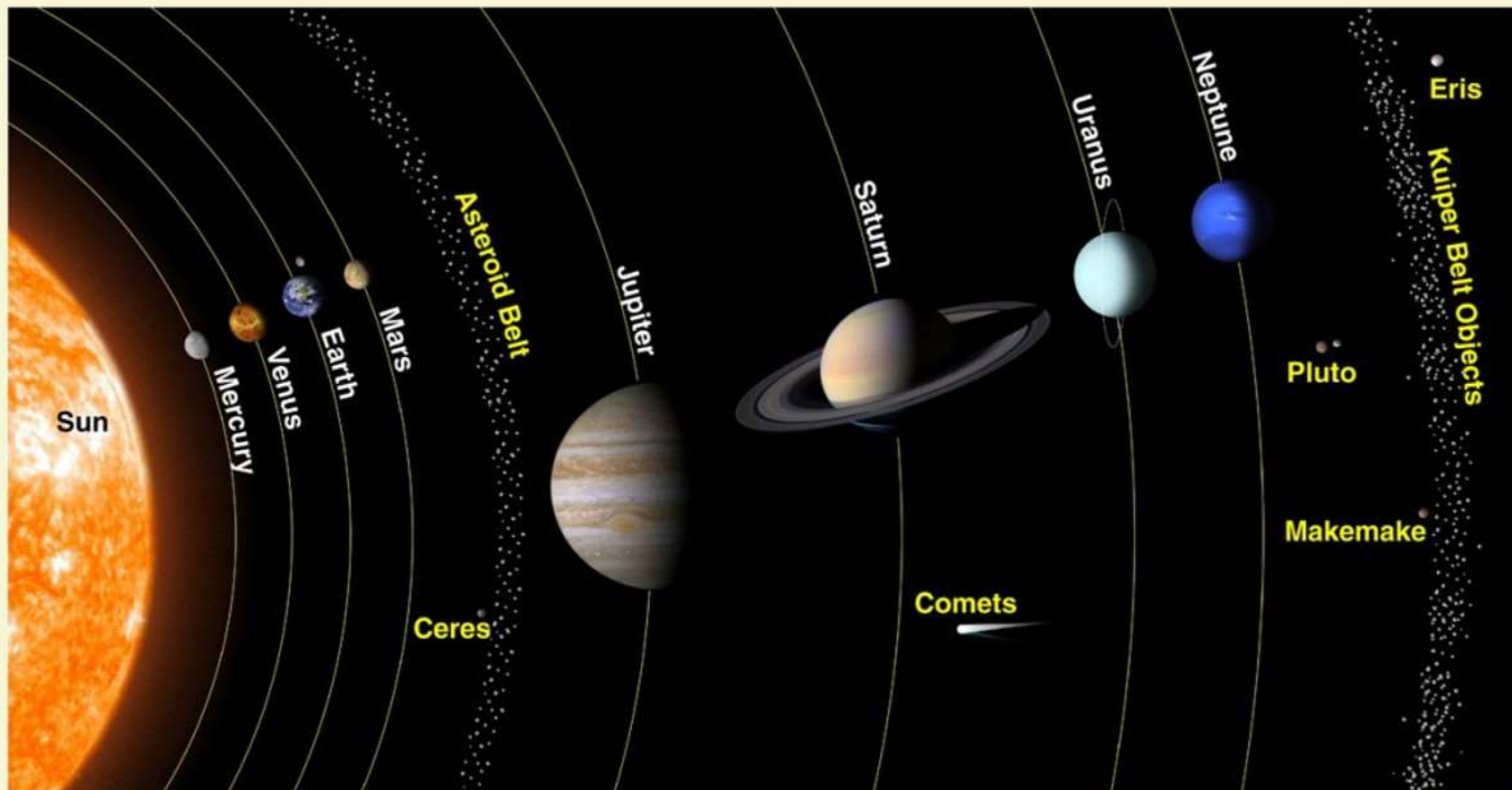
Deimos





Why Venus is hotter than Mercury?

- Venus is hotter because of its predominant CO_2 in the atmosphere.
 - It absorbs sunlight.
 - It creates a super greenhouse effect.
 - Mercury has no atmosphere.
 - It's temperature varies from 426°C and -173°C .
 - Average temperature for Venus is 462°C .
 - Venus has got thick atmosphere of CO_2 and this along with high pressure traps the heat.
 - Venus is often visible as the first bright star in the sky during twilight or the last bright star in the sky at dawn.
- As the Earth rotates from west to east, for an observer on the Earth, the Venus would rise before the Sun. For the Sun to rise, the Earth would have to rotate further by some degrees.
 - Prior to Sun appearing, Venus appears and this is known as 'Morning Star'.
 - In the evening, when the Sun sets, the Venus will still be above the observer's horizon and will be visible to the observer as 'Evening Star'.





Capella

Venus

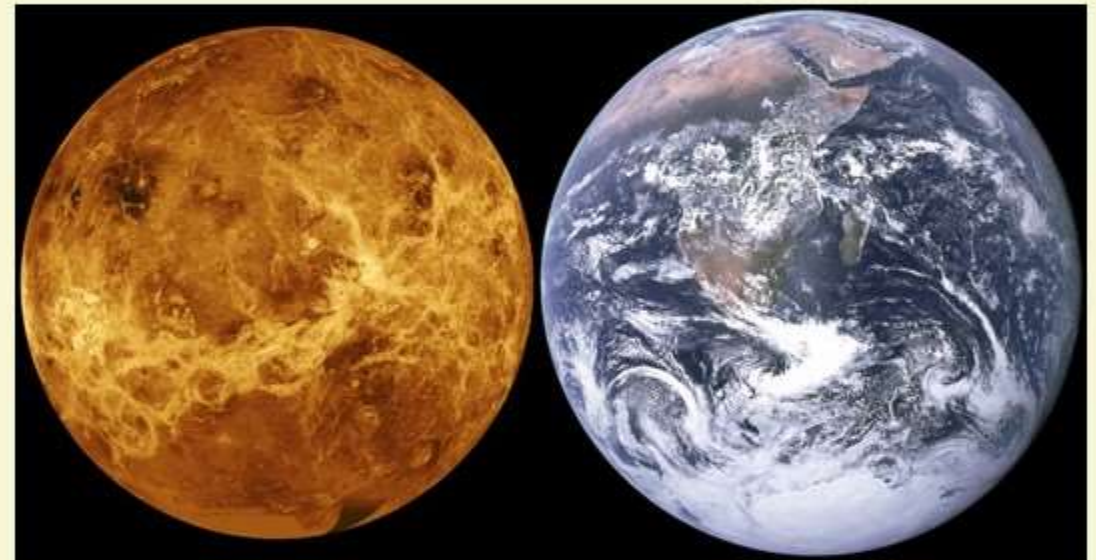
Mercury

E



Why Venus is called Earth's twin?

- The diameter of Venus is 95% of the Earth's diameter (Earth's diameter is 12,756 km).
- Venus has 90% of Earth's surface area, the mass is 81% of mass of Earth.
- Force of gravity is 90% of that on Earth.
- Both have metal cores surrounded by a mantle of silica rock.



However, Venus does not have magnetic field, it also does not have plate tectonics and it is too hot at around 462°C



Why Venus looks so bright?

- It is due to high albedo.
- Albedo is nothing but ratio of the reflected radiation to the incident radiation.
- It's value ranges from zero to one, zero for 'no reflection', one for 'perfect reflection'.
- It has albedo value of **0.75**, which is higher than most objects that are observable from Earth.
- So, that makes Venus the third brightest object in the sky preceded only by the Sun and moon.



- There are droplets of sulphuric acid and acidic crystals in the atmosphere of Venus.
- The smooth surfaces of these droplets reflect light very well.

High albedo value can also be attributed to the thick cover of reflective clouds enveloping its surface.



Why Mars is called Red Planet?

- It appears like a bright red star.
- The surface is rich in iron oxide and that's why, it looks red.
- Mars iron oxide would have formed a long time ago, when the planet had more liquid water.
- The rusty material was transported around the planet in dust clouds covering everything in a layer of dust.





Why Earth is called Blue Planet?

- The water bodies such as oceans, lakes, rivers appear blue.
- About 70% of the Earth's crust is covered with water.
- It produces blue colour, when viewed from space.





Why is the ocean looks blue?

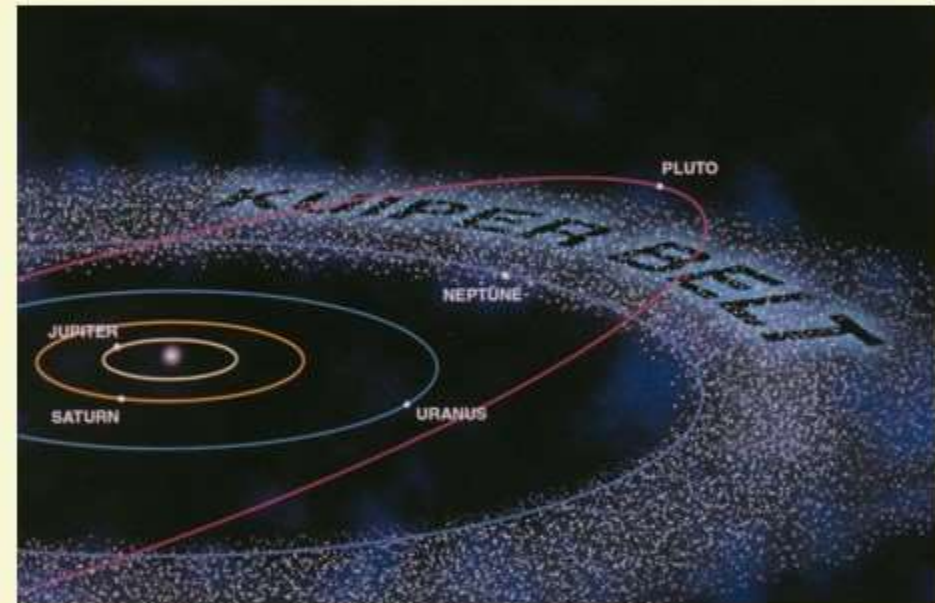
- Ocean is blue, because of the way it absorbs sunlight.
- When sunlight hits the ocean, the water strongly absorbs long wavelength colours at the red-end of the light spectrum as well as short-wavelength light, including violet and ultraviolet.
- The remaining light that we see is mostly made up of blue wavelengths.
- If the light bounces off the floating objects, such as sediment and algae sometimes, oceans may look red and green also.

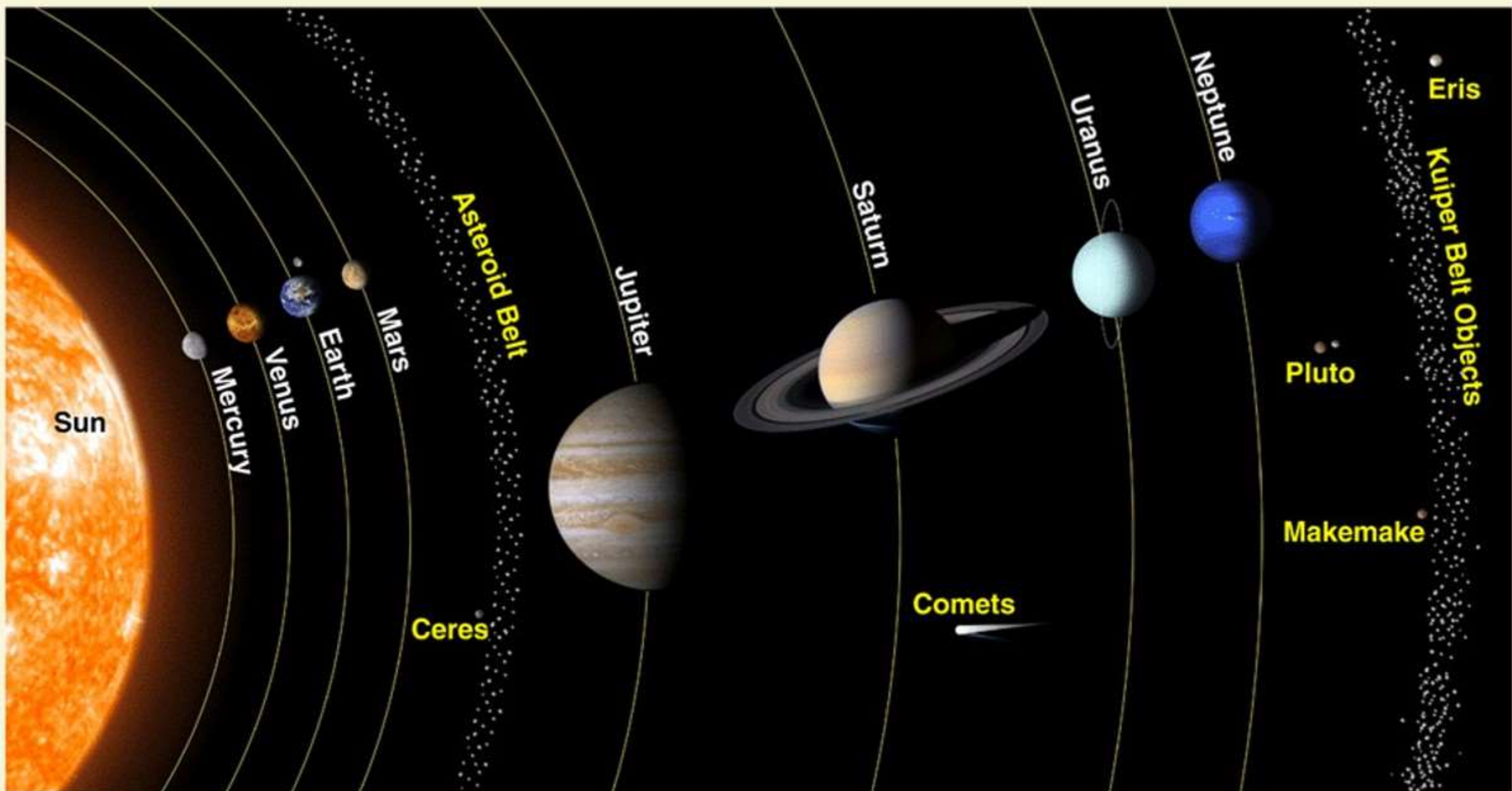




What do you understand by Kuiper Belt?

- It is the region of icy bodies, dwarf planets such as Pluto.
- It also includes comets, beyond the orbit of Neptune.
- Probably populated with hundreds of thousands of icy bodies larger than 100 km across and an estimated trillion or more comets.

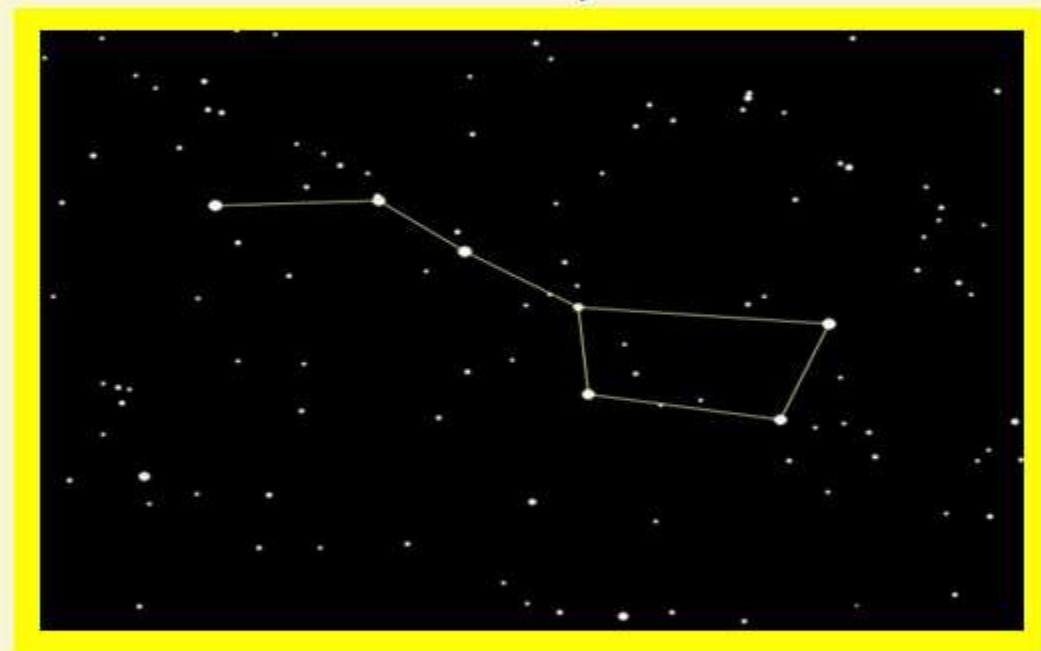






What do you understand by constellation of stars?

SAPTARISHI - Group of seven stars

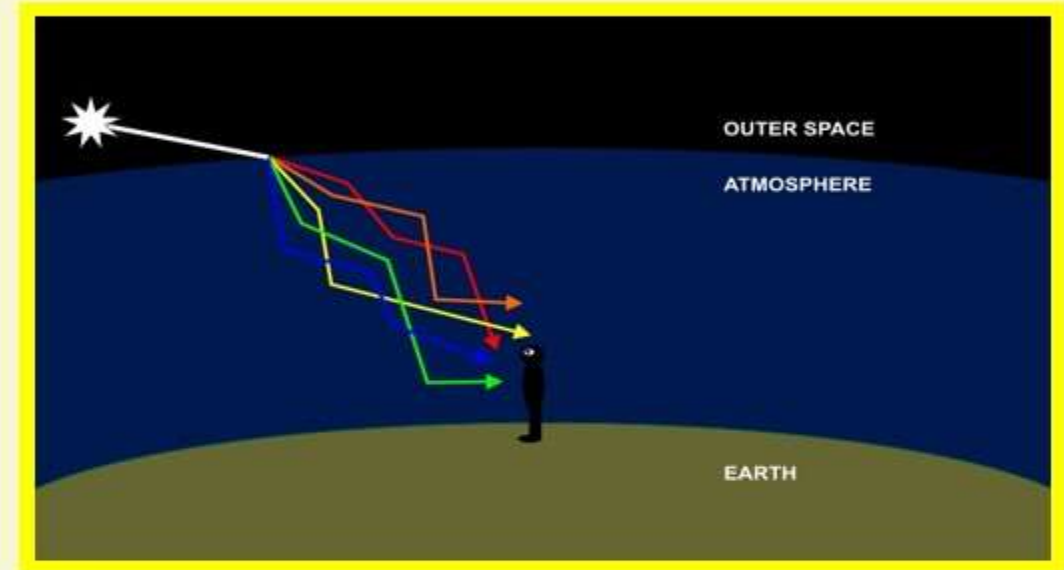


An easily recognized group of stars that appear to be located close together in the sky and that form a picture, if lines connecting them are imagined.



Why twinkling of stars occur?

- ✓ Stars twinkle because of turbulence in the atmosphere of the Earth.
- ✓ The star's light travels through many layers of the Earth's atmosphere.
- ✓ The light of the star is bent (refracted) many times and in random directions and this results in twinkling.
- ✓ The scientific name for the twinkling of stars is **stellar scintillation**.
- ✓ Stars closer to the horizon will appear to twinkle more than other stars.
- ✓ Stars would not appear to twinkle if we viewed them from outer space.



Planets do not usually twinkle, because they are so close to us; they appear big enough that the twinkling is not noticeable.

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